

ML101

Chapter 06: Linear Model Selection and Regularization

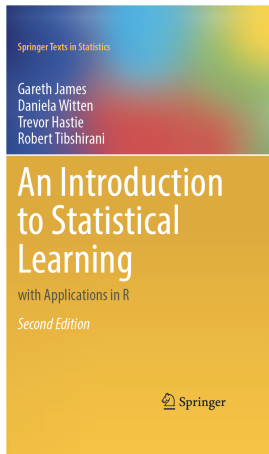
An Introduction to Statistical Learning, ISLRv2

James, Witten, Hastie & Tibshirani

September 3, 2026

An Introduction to Statistical Learning with Applications in R

a.k.a. ISLR



Content has been extracted from *An Introduction to Statistical Learning with Applications in R*, Second Edition, by James, Witten, Hastie & Tibshirani. Springer. 2023.
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In Praise of Linear Models!

- ▶ Recall the standard linear regression model:

$$Y = \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p + \epsilon$$

- ▶ **Why stick with linear models?**
 - ▶ High degree of **interpretability**.
 - ▶ Often competitive in **predictive performance** compared to non-linear black-box methods.
 - ▶ Computationally straightforward and well-understood theoretical foundations.
- ▶ In this lecture, we improve ordinary least squares (OLS) by considering alternative fitting procedures to overcome key limitations.

Why Consider Alternatives to Least Squares?

1. Prediction Accuracy

- ▶ When $n \gg p$, OLS estimates tend to have low variance.
- ▶ If n is not much larger than p , OLS fit can suffer from **high variability**, leading to overfitting.
- ▶ If $p > n$, OLS has **no unique solution** (variance is infinite).
- ▶ Constraining or shrinking coefficients trades a tiny increase in bias for a substantial drop in variance.

2. Model Interpretability

- ▶ Multiple regression models often include variables that are unrelated to the response.
- ▶ OLS rarely sets coefficient estimates exactly to zero ($\hat{\beta}_j \neq 0$).
- ▶ Retaining irrelevant variables introduces clutter and reduces interpretability.
- ▶ We desire automated methods for **feature selection** (sparsity).

Three Classes of Methods

1. Subset Selection:

- ▶ Identify a subset of the p predictors believed to be related to the response.
- ▶ Fit a model using standard least squares on the reduced set of variables.

2. Shrinkage (Regularization):

- ▶ Fit a model involving all p predictors, but shrink estimated coefficients towards zero relative to OLS.
- ▶ Reduces variance significantly; specific forms (e.g., Lasso) also perform variable selection.

3. Dimension Reduction:

- ▶ Project the p predictors into an M -dimensional subspace ($M < p$) using linear combinations.
- ▶ Fit a least squares regression model using these M new projected features.

Plan

Subset Selection

Shrinkage Methods

Dimension Reduction Methods

Considerations in High Dimensions

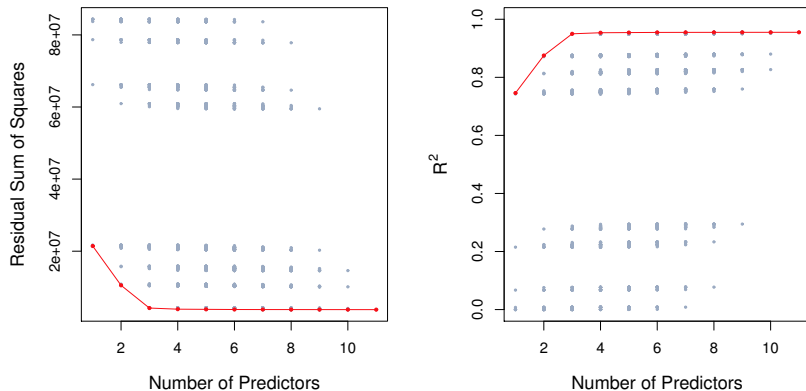
Summary

Best Subset Selection

Algorithm: Best Subset Selection

1. Let $\mathcal{M}_0$ denote the **null model**, which contains no predictors (predicts sample mean $\bar{y}$).
 2. For $k = 1, 2, \dots, p$:
 - (a) Fit all $\binom{p}{k} = \frac{p!}{k!(p-k)!}$ models containing exactly k predictors.
 - (b) Pick the best among these $\binom{p}{k}$ models, called $\mathcal{M}_k$ (lowest RSS or highest R^2).
 3. Select a single best model from among $\mathcal{M}_0, \mathcal{M}_1, \dots, \mathcal{M}_p$ using cross-validated prediction error, C_p (AIC), BIC, or adjusted R^2 .
- **Crucial distinction:** Step 2 uses training RSS/ R^2 to compare models of the *same size*. Step 3 uses test-error estimates to compare models of *different sizes*.

Best Subset Selection on Credit Data



- ▶ **Left Panel:** Residual Sum of Squares (RSS) for all possible subset models of the **Credit** data. As subset size k increases, training RSS decreases monotonically.
- ▶ **Right Panel:** Corresponding R^2 values increasing monotonically to 1.
- ▶ **Red Frontier:** Highlights the best model $\mathcal{M}_k$ for each subset size k .

Extension to Other Models

- ▶ Although we have presented best subset selection here for least squares regression, the same ideas apply to other types of models, such as logistic regression.
- ▶ The *deviance* –negative two times the maximized log-likelihood– plays the role of RSS for a broader class of models.

Stepwise Selection

- For computational reasons, best subset selection cannot be applied with very large p . **Why not?**

Stepwise Selection

- ▶ For computational reasons, best subset selection cannot be applied with very large p . **Why not?** $2^p \implies 2^{10} \approx 1000$ but $2^{40} \approx 1.1 \times 10^{12}$
- ▶ Best subset selection may also suffer from statistical problems when p is large: larger the search space, the higher the chance of finding models that look good on the training data, even though they might not have any predictive power on future data.

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- ▶ Thus an enormous search space can lead to **overfitting** and high variance of the coefficient estimates (we could find an extremely good model in the training data by chance, some combination of noise features will align with the training response simply by random chance).
- ▶ For both of these reasons, **stepwise** methods, which explore a far more restricted set of models, are attractive alternatives to best subset selection.

Forward Stepwise Selection

- ▶ Best subset selection is computationally intractable for large p .
- ▶ **Forward stepwise selection** offers a greedy, computationally efficient alternative:

Algorithm: Forward Stepwise Selection

1. Let $\mathcal{M}_0$ denote the null model.
 2. For $k = 0, 1, \dots, p - 1$:
 - (a) Consider all $p - k$ models that augment $\mathcal{M}_k$ with one additional predictor.
 - (b) Choose the **best** among these $p - k$ models (smallest RSS / highest R^2), and call it $\mathcal{M}_{k+1}$.
 3. Select the overall best model among $\mathcal{M}_0, \dots, \mathcal{M}_p$ via CV, C_p , AIC, BIC, or adjusted R^2 .
- ▶ Fits only $1 + \sum_{k=0}^{p-1} (p - k) = 1 + \frac{p(p+1)}{2} \approx p^2$ models (for $p = 20$: 211 vs. 1,048,576).

Backward Stepwise Selection

Algorithm: Backward Stepwise Selection

1. Let $\mathcal{M}_p$ denote the **full model**, which contains all p predictors.
 2. For $k = p, p - 1, \dots, 1$:
 - (a) Consider all k models that contain all but one predictor in $\mathcal{M}_k$ ($k - 1$ predictors).
 - (b) Choose the **best** among these k models (smallest RSS / highest R^2), and call it $\mathcal{M}_{k-1}$.
 3. Select the single best model among $\mathcal{M}_0, \dots, \mathcal{M}_p$ using CV, C_p , AIC, BIC, or adjusted R^2 .
- ▶ Like forward stepwise, considers only $1 + \frac{p(p+1)}{2} \approx p^2$ models.
 - ▶ **Constraint:** Requires $n > p$ so that the initial full model $\mathcal{M}_p$ can be fit by least squares.
 - ▶ In contrast, forward stepwise can be used when $n < p$ (constructing submodels up to $\mathcal{M}_{n-1}$).

Credit Data: Subset Comparison

# Variables	Best Subset Selection	Forward Stepwise Selection
1	rating	rating
2	rating, income	rating, income
3	rating, income, student	rating, income, student
4	cards, income, student, limit	rating, income, student, limit

- ▶ For $k = 1, 2, 3$, both methods identify the exact same predictor set.
- ▶ At $k = 4$, best subset drops **rating** in favor of **cards** and **limit**.
- ▶ Forward stepwise is **nested** ($\mathcal{M}_k \subset \mathcal{M}_{k+1}$); it cannot drop **rating** once added.

Choosing the Optimal Model: Adjusted Criteria

To compare models with different numbers of predictors d , we adjust training RSS:

Mallows' C_p and Akaike Information Criteria (AIC)

$$C_p = \frac{1}{n} (\text{RSS} + 2d\hat{\sigma}^2)$$

$$AIC = -2 \log L + 2 \cdot d$$

- ▶ where L is the *maximized value of the likelihood* function.
- ▶ $\hat{\sigma}^2$ is an unbiased estimate of error variance σ^2 from the full model.
- ▶ Penalizes model size by adding $2d\hat{\sigma}^2$.

Bayesian Information Criteria (BIC) and Adjusted R^2

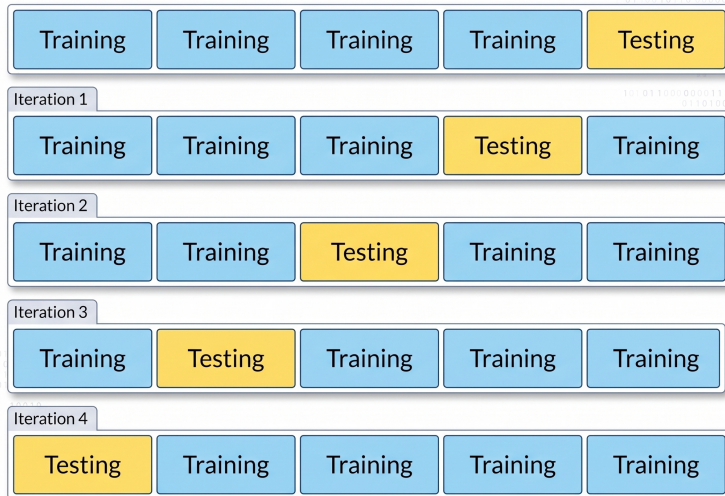
$$BIC = \frac{1}{n} (\text{RSS} + \log(n)d\hat{\sigma}^2)$$

$$\text{Adjusted } R^2 = 1 - \frac{\text{RSS}/(n - d - 1)}{\text{TSS}/(n - 1)}$$

- ▶ For $n > 7$, $\log(n) > 2$; hence **BIC penalizes larger models more heavily** than C_p .
- ▶ For C_p , AIC, BIC: **smaller is better**.
- ▶ For Adjusted R^2 : **larger is better**.

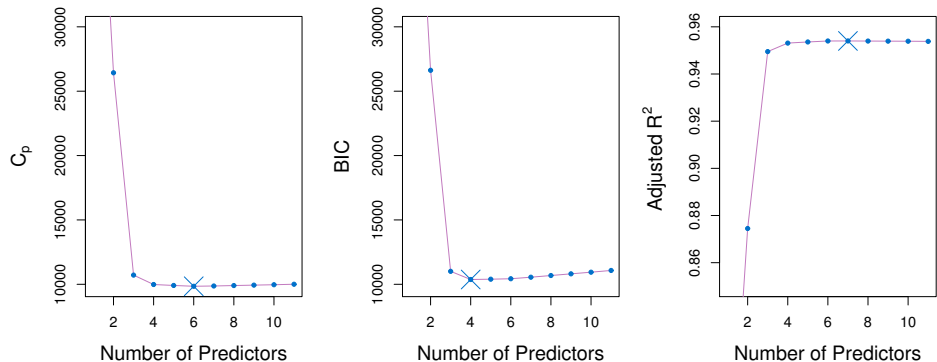
Cross-validation

5 Fold Cross-Validation



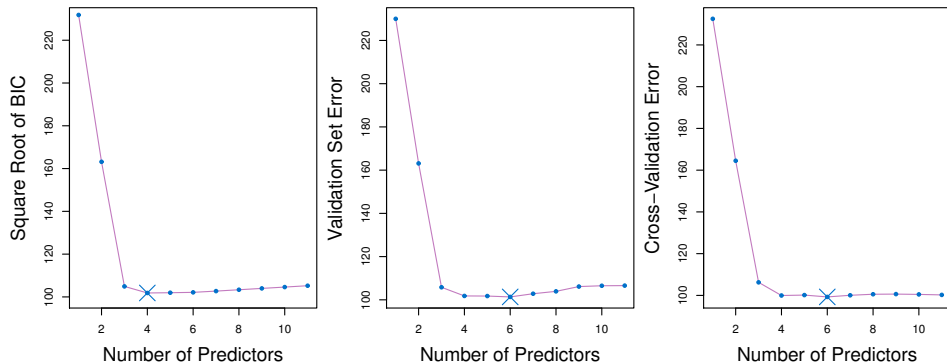
Total Dataset: 5 Folds. In each iteration, one unique fold is used for testing, and the rest for training. Results are averaged.

Model Selection: Adjusted Criteria vs. Validation/CV



- Plots of C_p , BIC , and Adjusted R^2 for **Credit** data models.
- BIC chooses a parsimonious 4-variable model (**income**, **limit**, **cards**, **student**), while C_p and Adjusted R^2 choose 6 variables.

Model Selection: Adjusted Criteria vs. Validation/CV –continued



- Cross-validation curve is flat between 4 and 11 variables, showing that extra variables add little test predictive power.

The One-Standard-Error Rule

Definition: One-Standard-Error Rule

When comparing a set of models via cross-validation, calculate the standard error of the estimated test MSE for each model size.

- ▶ Identify the model with the minimum cross-validation error.
- ▶ Select the **simplest (most parsimonious) model** whose CV error is within **1 standard error** of the minimum.
- ▶ **Rationale:** Acknowledges that differences in CV error within 1 SE are likely due to random sampling variability.
- ▶ Adheres to Occam's razor: favor simpler, more interpretable models when empirical performance is statistically indistinguishable.

Plan

Subset Selection

Shrinkage Methods

Dimension Reduction Methods

Considerations in High Dimensions

Summary

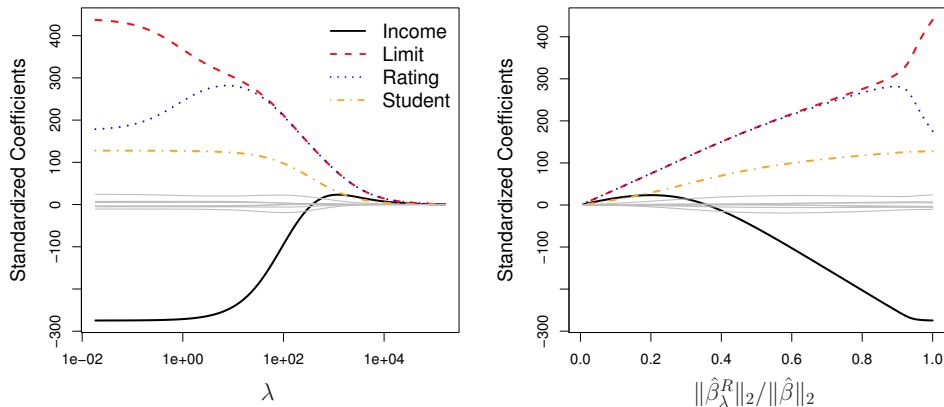
Ridge Regression

- ▶ Standard OLS minimizes $\text{RSS} = \sum_{i=1}^n \left(y_i - \beta_0 - \sum_{j=1}^p \beta_j x_{ij} \right)^2$.
- ▶ **Ridge regression** minimizes a penalized objective:

$$\hat{\beta}_{\lambda}^R = \arg \min_{\beta} \left\{ \sum_{i=1}^n \left(y_i - \beta_0 - \sum_{j=1}^p \beta_j x_{ij} \right)^2 + \lambda \sum_{j=1}^p \beta_j^2 \right\} = \arg \min_{\beta} \{ \text{RSS} + \lambda \|\beta\|_2^2 \}$$

- ▶ $\lambda \geq 0$ is a **tuning parameter**:
 - ▶ When $\lambda = 0$: penalty has no effect $\implies \hat{\beta}_0^R = \hat{\beta}^{\text{OLS}}$.
 - ▶ As $\lambda \rightarrow \infty$: shrinkage penalty dominates $\implies \hat{\beta}_{\lambda}^R \rightarrow 0$.
- ▶ The intercept β_0 is *not* penalized (estimated simply as $\bar{y}$ when predictors are centered).
- ▶ Uses the squared ℓ_2 penalty: $\|\beta\|_2 = \sqrt{\sum_{j=1}^p \beta_j^2}$.

Ridge Regression: Coefficient Profiles on Credit Data



- ▶ **Left Panel:** Standardized ridge regression coefficients for **Credit** predictors plotted against λ . As λ increases, all coefficients shrink continuously towards zero.
- ▶ **Right Panel:** The same coefficients plotted as a function of $\|\hat{\beta}_\lambda^R\|_2 / \|\hat{\beta}^{\text{OLS}}\|_2$ (amount of shrinkage relative to OLS).
- ▶ Notice that **income**, **limit**, **rating**, and **student** remain the largest throughout.

Ridge Regression: Scaling of Predictors

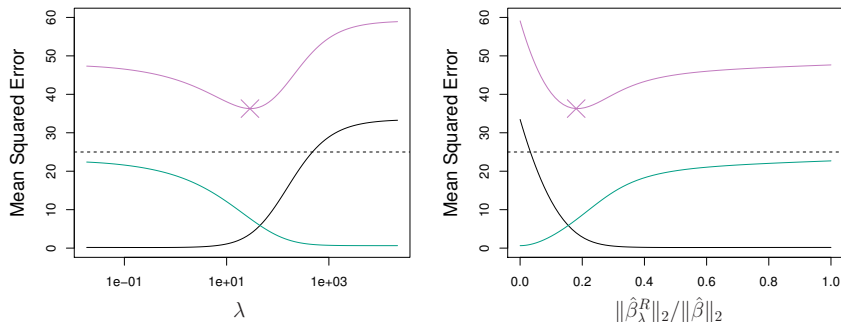
- ▶ OLS coefficient estimates are **scale equivariant**: multiplying X_j by a constant c simply scales $\hat{\beta}_j$ by $1/c$, leaving $X_j\hat{\beta}_j$ unchanged.
- ▶ In contrast, the ridge penalty $\lambda \sum \beta_j^2$ is **sensitive to predictor scaling**!
- ▶ A variable measured in dollars will have much smaller coefficients (and hence smaller penalty) than the same variable measured in thousands of dollars.

Standardization Formula

Before applying ridge regression, standardizing all predictors to unit variance is essential:

$$\tilde{x}_{ij} = \frac{x_{ij}}{\sqrt{\frac{1}{n} \sum_{i=1}^n (x_{ij} - \bar{x}_j)^2}}$$

Why Does Ridge Regression Improve Over OLS?



- ▶ Simulation setting with $p = 45$ predictors and $n = 50$ observations.
- ▶ **Variance (green curve):** Extremely high at $\lambda = 0$ (OLS), but drops dramatically as λ increases.
- ▶ **Squared Bias (black curve):** Zero at $\lambda = 0$, increases slowly at first, then climbs rapidly.
- ▶ **Test MSE (purple curve):** Exhibits a clear U-shape, achieving a minimum at an intermediate value of $\lambda \approx 30$.

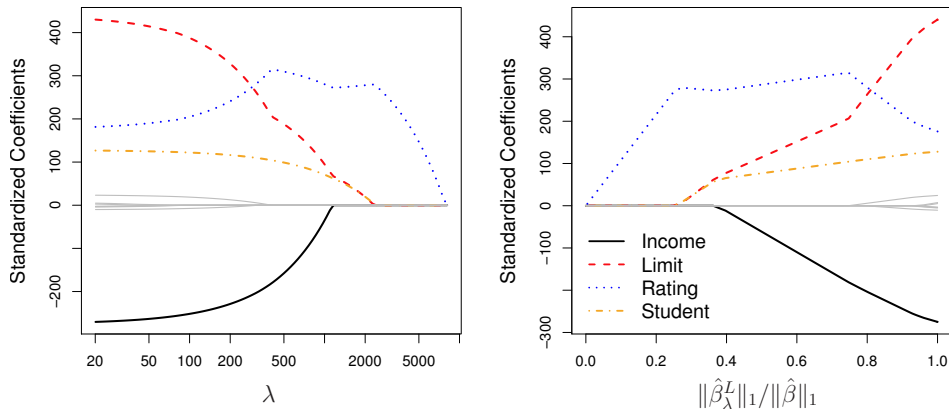
The Lasso

- ▶ **Disadvantage of Ridge:** It will shrink all coefficients close to zero, but *never sets any coefficient exactly to zero* (unless $\lambda = \infty$).
- ▶ **The Lasso** replaces the ℓ_2 penalty with an ℓ_1 penalty:

$$\hat{\beta}_{\lambda}^L = \arg \min_{\beta} \left\{ \sum_{i=1}^n \left(y_i - \beta_0 - \sum_{j=1}^p \beta_j x_{ij} \right)^2 + \lambda \sum_{j=1}^p |\beta_j| \right\} = \arg \min_{\beta} \{ \text{RSS} + \lambda \|\beta\|_1 \}$$

- ▶ $\|\beta\|_1 = \sum_{j=1}^p |\beta_j|$ is the ℓ_1 norm.
- ▶ The ℓ_1 penalty forces some coefficients to be **identically zero** when λ is sufficiently large.
- ▶ Hence, Lasso yields **sparse models** and performs automatic variable selection!

Example: Credit Dataset (Lasso)



- ▶ **Left Panel:** Standardized Lasso coefficients plotted against λ . For large λ , all coefficients are 0. As λ decreases, **rating** enters first, followed by **student**, **limit**, and **income**.
- ▶ **Right Panel:** Lasso coefficients plotted against $\|\hat{\beta}_\lambda^L\|_1 / \|\hat{\beta}^{\text{OLS}}\|_1$. Clear variable entry points demonstrate step-by-step feature inclusion.

Budget Constraint Formulations

The Lasso and Ridge problems can be equivalently formulated as constrained optimizations:

Lasso Optimization (ℓ_1 Budget)

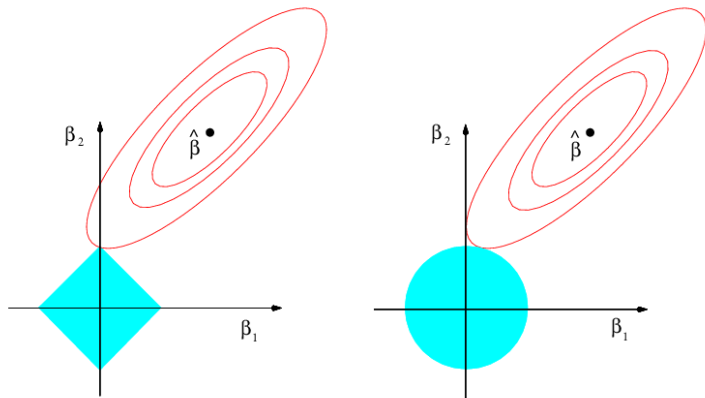
Ridge Optimization (ℓ_2 Budget)

$$\min_{\beta} \text{RSS} \quad \text{subject to} \quad \sum_{j=1}^p |\beta_j| \leq s$$

$$\min_{\beta} \text{RSS} \quad \text{subject to} \quad \sum_{j=1}^p \beta_j^2 \leq s$$

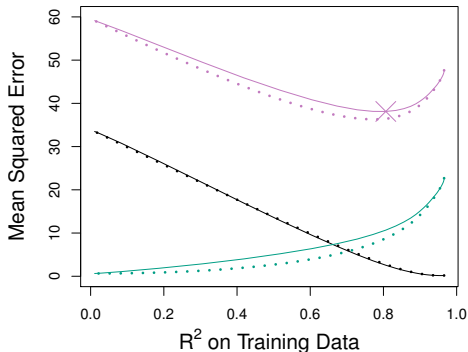
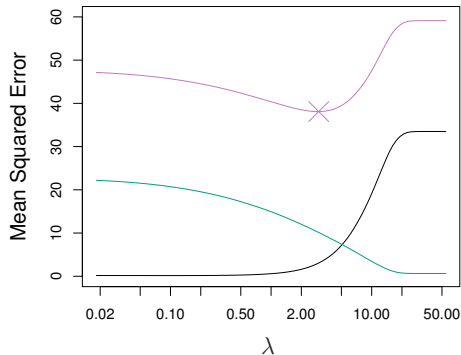
- ▶ For every λ , there corresponds a budget $s \geq 0$.
- ▶ If s is very large, the budget constraint is non-binding, yielding the unconstrained OLS solution.
- ▶ If s is small, the coefficients are constrained to lie strictly within the budget region.

The Variable Selection Property of the Lasso



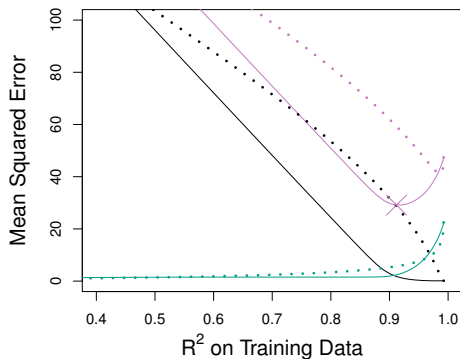
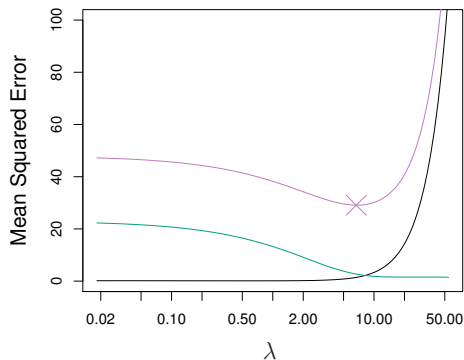
- ▶ **Lasso (Left):** The constraint region is a diamond ($|\beta_1| + |\beta_2| \leq s$) with sharp corners on the coordinate axes. The elliptical RSS contours frequently contact the diamond at a corner, forcing $\beta_1 = 0$.
- ▶ **Ridge (Right):** The constraint region is a circle ($\beta_1^2 + \beta_2^2 \leq s$) with no corners. Contact points almost never lie on the axes, so both $\beta_1, \beta_2 \neq 0$.

Comparing Lasso and Ridge Regression



- ▶ **Left:** Plots of squared bias (black), variance (green), and test MSE (purple) for the lasso on simulated data set ($n = 50$ and $p = 45$).
- ▶ **Right:** Comparison of squared bias, variance and test MSE between lasso (solid) and ridge (dashed). Both are plotted against their R^2 on the training data, as a common form of indexing. The crosses in both plots indicate the lasso model for which the MSE is smallest.

Comparing Lasso and Ridge Regression



- ▶ **Left:** Plots of squared bias (black), variance (green), and test *MSE* (purple) for the lasso. The simulated data is similar to that in previous slide, except that now only two predictors are related to the response.
- ▶ **Right:** Comparison of squared bias, variance and test *MSE* between lasso (solid) and ridge (dashed). Both are plotted against their R^2 on the training data, as a common form of indexing. The crosses in both plots indicate the lasso model for which the *MSE* is smallest.

Comparing Lasso and Ridge Regression –continued

- ▶ **Dense Setting (All 45 predictors non-zero):** Ridge regression slightly outperforms Lasso in test MSE because shrinking all variables together matches the underlying truth.
- ▶ **Sparse Setting (Only 2 of 45 predictors non-zero):** Lasso significantly outperforms Ridge because it successfully zeros out the 43 noise variables, drastically cutting variance.

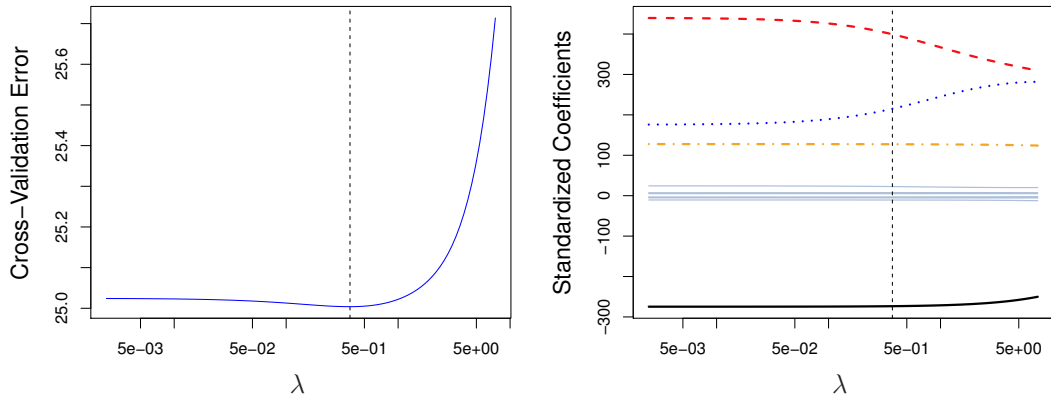
Conclusions

- ▶ These two examples illustrate that neither ridge regression nor the lasso will universally dominate the other.
- ▶ In general, one might expect the lasso to perform better when the response is a function of only a relatively small number of predictors.
- ▶ However, the number of predictors that is related to the response is never known **a priori** for real data sets.
- ▶ A technique such as cross-validation can be used in order to determine which approach is better on a particular data set.

Selecting the Tuning Parameter for Ridge Regression and Lasso

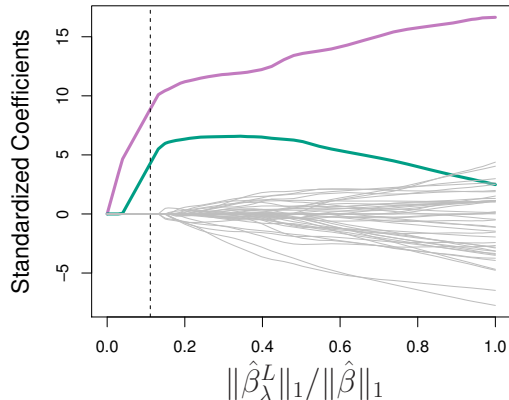
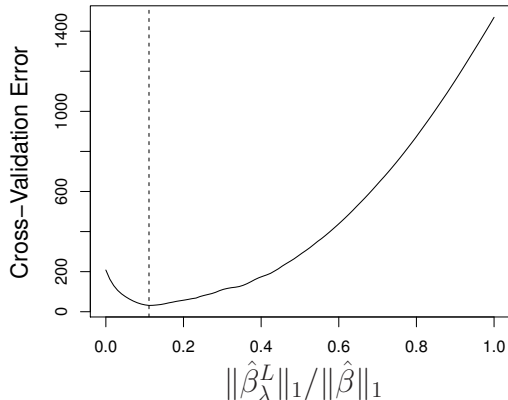
- ▶ As for subset selection, for ridge regression and lasso we require a method to determine which of the models under consideration is best.
- ▶ That is, we require a method selecting a value for the tuning parameter λ or equivalently, the value of the constraint s .
- ▶ **Cross-validation** provides a simple way to tackle this problem. We choose a grid of λ values, and compute the cross-validation error rate for each value of λ .
- ▶ We then select the tuning parameter value for which the cross-validation error is smallest.
- ▶ Finally, the model is re-fit using all of the available observations and the selected value of the tuning parameter.

Selecting the Regularization Parameter λ – Credit data example



- ▶ **Left Panel:** Cross-validation errors that result from applying ridge regression to the Credit data set with various values of λ .
- ▶ **Right Panel:** The coefficient estimates as a function of λ . The vertical dashed lines indicates the value of λ selected by cross-validation.

Simulated data example



- ▶ **Left Panel:** Ten-fold cross-validation MSE for the lasso, applied to the sparse simulated data set from Slide 39.
- ▶ **Right Panel:** The corresponding lasso coefficient estimates are displayed. The vertical dashed lines indicate the lasso fit for which the cross-validation error is smallest.

Plan

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Summary

Dimension Reduction Methods

- ▶ Rather than selecting subsets or shrinking original features, **dimension reduction methods** transform predictors:

$$Z_m = \sum_{j=1}^p \phi_{jm} X_j, \quad m = 1, \dots, M \quad (M < p)$$

- ▶ Then fit an OLS regression model using the M transformed features:

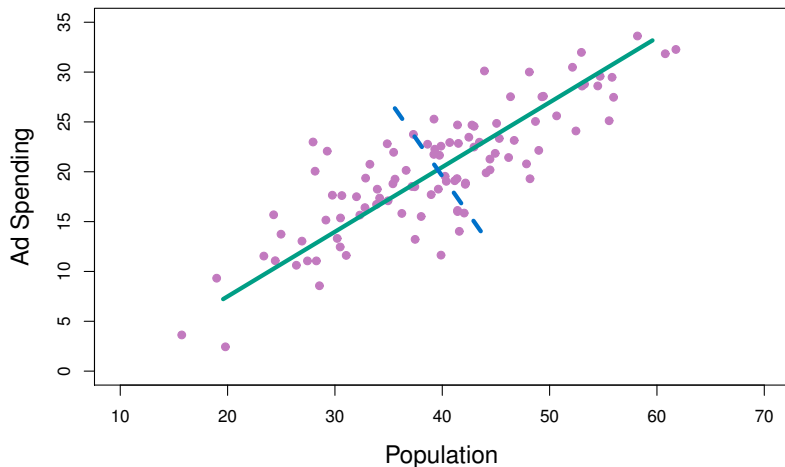
$$y_i = \theta_0 + \sum_{m=1}^M \theta_m z_{im} + \epsilon_i, \quad i = 1, \dots, n$$

- ▶ Notice that $\sum_{m=1}^M \theta_m z_{im} = \sum_{j=1}^p \left(\sum_{m=1}^M \theta_m \phi_{jm} \right) x_{ij} = \sum_{j=1}^p \beta_j x_{ij}$, where $\beta_j = \sum_{m=1}^M \theta_m \phi_{jm}$.
- ▶ Constrains coefficient vector to an M -dimensional subspace, mitigating overfitting.

Principal Components Regression (PCR)

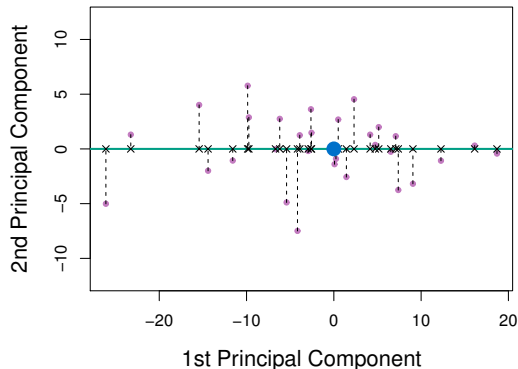
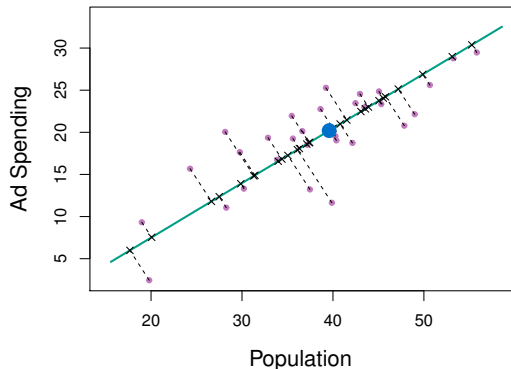
- ▶ **Principal Components Regression (PCR)** constructs the first M principal components $Z_1, \dots, Z_M$ and uses them as predictors in least squares regression.
- ▶ **Key Assumption:** Directions in feature space that capture the greatest variance in $\mathbf{X}$ are also the directions most associated with Y .
- ▶ **Properties of PCR:**
 - ▶ PCR is **unsupervised**: response Y is *not* used to determine components.
 - ▶ PCR is *not* a feature selection method: each Z_m is a linear combination of all p original features.
 - ▶ Always **standardize predictors** prior to computing principal components.
 - ▶ Tuning parameter M ($1 \leq M \leq p$) is chosen via cross-validation.

Pictures of PCA



- The population size (**pop**) and ad spending (**ad**) for 100 different cities are shown as purple circles. The green solid line indicates the first principal component, and the blue dashed line indicates the second principal component.

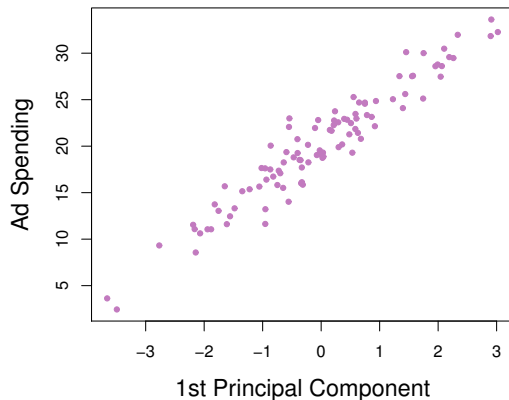
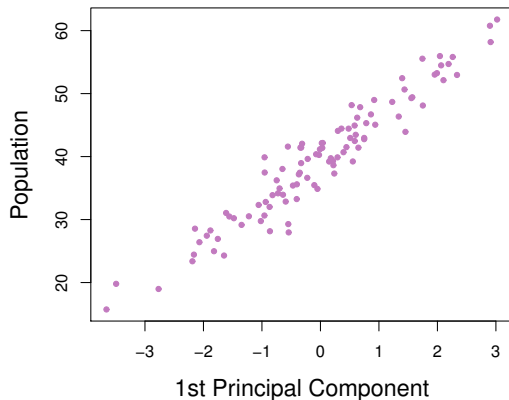
Pictures of PCA –continued



A subset of the advertising data.

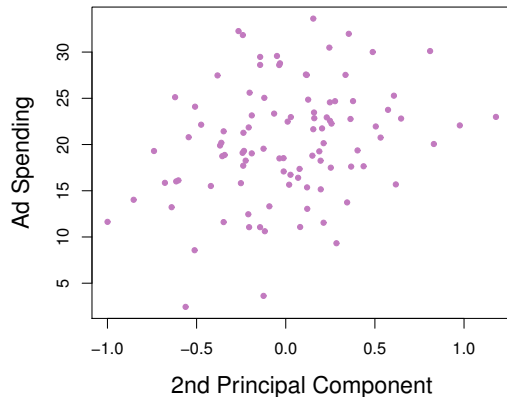
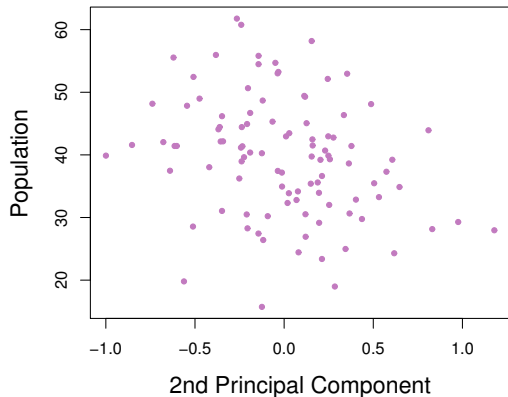
- ▶ **Left:** The first principal component, chosen to minimize the sum of the squared perpendicular distances to each point, is shown in green. These distances are represented using the black dashed line segments.
- ▶ **Right:** The left-hand panel has been rotated so that the first principal component lies on the x-axis.

Pictures of PCA –continued



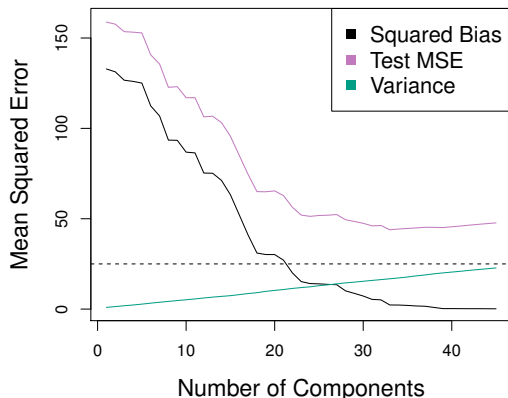
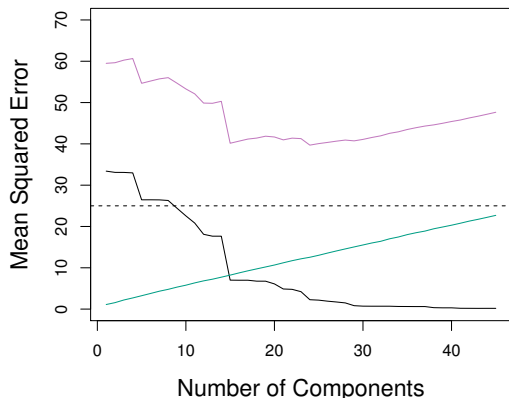
- Plots of the first principal component scores z_{i1} versus **pop** and **ad**. The relationships are strong.

Pictures of PCA –continued



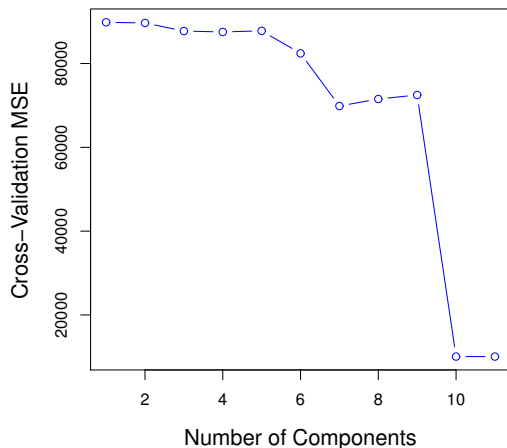
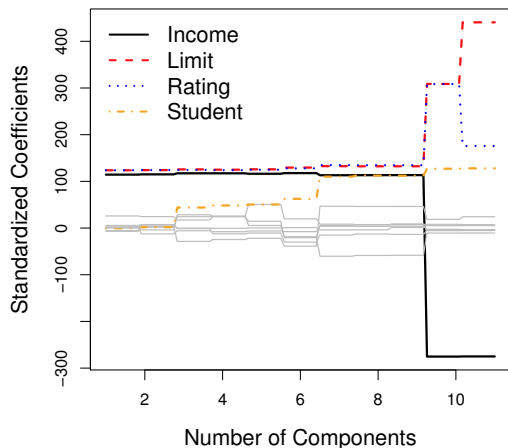
- Plots of the second principal component scores z_{i2} versus **pop** and **ad**. The relationships are weak.

Application to Principal Components Regression



- ▶ *PCR* was applied to two simulated data sets. The black, green, and purple lines correspond to squared bias, variance, and test mean squared error, respectively.
- ▶ **Left:** Simulated data from slide 32.
- ▶ **Right:** Simulated data from slide 39.

Choosing the number of directions M



- ▶ **Left:** *PCR* standardized coefficient estimates on the **Credit** data set for different values of M .
- ▶ **Right:** The 10-fold cross validation *MSE* obtained using *PCR*, as a function of M .

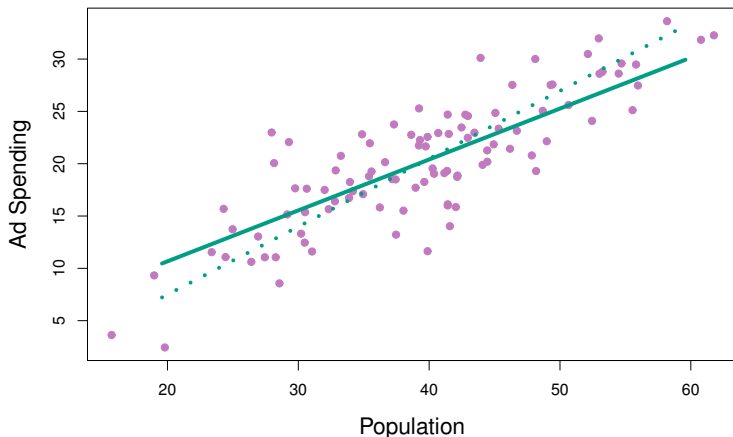
Partial Least Squares (PLS)

- ▶ **Drawback of PCR:** As an unsupervised method, PCA may construct directions with large predictor variance that happen to have zero correlation with Y .
- ▶ **Partial Least Squares (PLS)** is a **supervised** dimension reduction technique:

Algorithm: Partial Least Squares (First Directions)

1. Standardize all p predictors and center the response Y .
2. Compute the first direction $Z_1 = \sum_{j=1}^p \phi_{1j} X_j$ by setting ϕ_{1j} equal to the slope from regressing Y onto X_j ($\phi_{1j} \propto \text{Cor}(X_j, Y)$).
3. **Orthogonalize:** Regress each X_j on Z_1 and compute residuals (unexplained variation).
4. Compute Z_2 on these residuals using the same strategy, and repeat for M steps.
5. Fit multiple linear regression of Y on $Z_1, \dots, Z_M$ via least squares.

PLS vs. PCR Comparison



- ▶ Comparison of first principal component (unsupervised PCR direction, dotted line) and first partial least squares direction (supervised PLS direction, solid line) on advertising data.
- ▶ PLS tilts the direction toward **pop** because **pop** is more strongly correlated with the sales response than **ad** spending.

Plan

Subset Selection

Shrinkage Methods

Dimension Reduction Methods

Considerations in High Dimensions

Summary

High-Dimensional Data: What Goes Wrong?

- ▶ Modern datasets frequently operate in the **high-dimensional regime** ($p > n$ or $p \approx n$).
- ▶ Examples: Genomics (thousands of SNPs, hundreds of patients), text processing, finance.

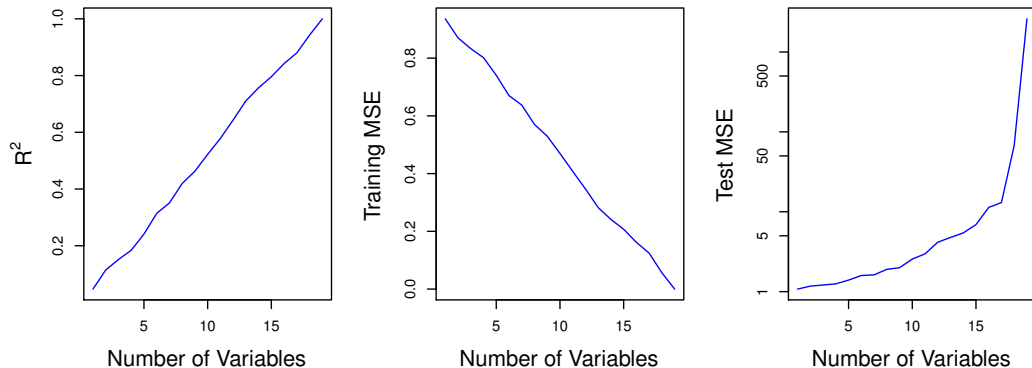
Pitfalls of OLS when $p \geq n$

- ▶ OLS will fit the training data perfectly ($R^2 = 1$, $\text{RSS} = 0$).
- ▶ Infinitely many solutions exist; standard error formulas fail ($\hat{\sigma}^2 = 0$).
- ▶ Severe **overfitting** causes catastrophic test set error.

Failure of Classical Criteria

- ▶ C_p , AIC, and BIC require estimating σ^2 from the full model, which is impossible when $p > n$.
- ▶ Adjusted R^2 easily produces artificial values of 1.
- ▶ Regularization or cross-validation is strictly mandatory.

High Dimensions: Training vs. Test Performance



- ▶ Simulated experiment with fixed sample size $n = 20$ where unrelated noise features are added one by one.
- ▶ **Left:** R^2 climbs to 1.0 as $p \rightarrow 20$.
- ▶ **Center:** Training MSE drops to 0.0.
- ▶ **Right:** Test set MSE explodes as additional noise predictors enter the model.

Interpreting Results in High Dimensions

- ▶ **Extreme Collinearity:** In high dimensions, any variable can typically be written as an almost exact linear combination of the others.
- ▶ **Never Claim "The" True Model:**
 - ▶ If Lasso or Forward Stepwise selects 15 genes to predict cancer recurrence, it does *not* mean those 15 genes are the only causal drivers.
 - ▶ Other correlated gene sets would achieve nearly identical predictive accuracy.
- ▶ **Validation is Mandatory:**
 - ▶ Never trust training set performance or standard p -values after model selection.
 - ▶ Model assessment must be conducted on completely independent test data or through rigorous cross-validation.

Plan

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Summary

Summary

Method	Selection Type	Sparsity	Key Strength / When to Use
Best Subset	Discrete Search	Yes	Optimal for very small p ($p \leq 20$).
Forward Stepwise	Greedy Search	Yes	Scalable subset method; works when $p > n$.
Ridge Regression	ℓ_2 Shrinkage	No	Superb under multicollinearity & dense signals.
The Lasso	ℓ_1 Shrinkage	Yes	Produces sparse, interpretable models.
PCR	Unsupervised Subspace	No	High collinearity; large predictor variance.
PLS	Supervised Subspace	No	High collinearity; supervised direction tuning.

Summary

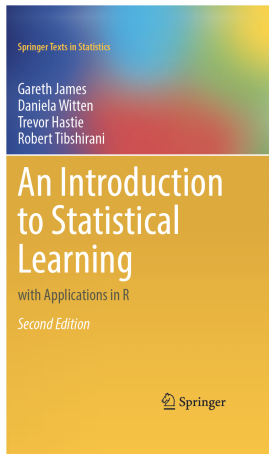
Core Conclusion

No single method uniformly dominates. When the true model is sparse, Lasso and subset selection excel; when many predictors have small, non-zero effects, Ridge regression and PCR often perform best. Tuning parameters (λ, M) should always be chosen via cross-validation.

End of Chapter 6.

An Introduction to Statistical Learning with Applications in R

a.k.a. ISLR



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